Droplet sedimentation, precipitation suppression and resolution in the Nd–LWP response of DP-EAMxx (DYCOMS-II RF02)

Technical report, version 4

Prepared by Claude Code (AI assistant) on behalf of Naser Mahfouz
for Naser Mahfouz and Johannes Mülmenstädt

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Abstract

We ran DP-EAMxx on the DYCOMS-II RF02 nocturnal drizzling stratocumulus for 12 h. The matrix covers grid spacing Δx from 40 km to 312 m, droplet number Nd from 20 to 1000 cm^{-3}, a cloud-droplet sedimentation multiplier ($\times$0.5, 1, 2), precipitation suppression (KK2000 Nd dependence on or off) and, below 1 km, 3D turbulence on or off. In all, 340 runs are complete and analysed. That includes all 36 runs at 312 m with 3D turbulence off, and 16 of the 36 with it on (76 % of 312 m model-hours). Section 3 adds process-level diagnostics. The Nd–LWP response depends strongly on resolution, and the two microphysical pathways trade places as Δx shrinks. At $\geq$10 km, LWP rises monotonically with Nd, and precipitation suppression explains all of that rise. As Δx shrinks, entrainment increases and depends more strongly on Nd, the precipitation-suppression gain collapses (from +134 $\mathrm{g\,m}^{-2}$ at 40 km to +6 $\mathrm{g\,m}^{-2}$ at 312 m, both at Nd 1000), and the LWP maximum moves to lower Nd (Nd > 1000 at $\geq$10 km, 500 at 5 km, 200 at 2.5 km–625 m, 100 at 312 m). The result is an inverted V whose falling branch is entrainment-driven: it persists with precipitation suppression off, and faster sedimentation strengthens it. The low-Nd branch ($S_{20\to50}$) barely changes with resolution below 2.5 km, while LWP itself has not converged at 312 m. 3D turbulence changes LWP by less than 1.5 $\mathrm{g\,m}^{-2}$ at both 625 m and 312 m.

Key results

1. **The inverted V forms and its peak moves to lower Nd as Δx shrinks** (Figs. 1, 2; Table 1). With sed$\times$1 and suppression on, LWP peaks at Nd $\geq$1000 at $\geq$10 km, at 500 at 5 km, at 200 from 2.5 km to 625 m, and at 100 at 312 m. The high-Nd susceptibility $S_{100\to200}$ falls monotonically: 0.27 (40 km), 0.20 (5 km), 0.11 (2.5 km), 0.07 (1.25 km), 0.03 (625 m), -0.05 (312 m) (Fig. 3).
2. **Precipitation suppression fades with resolution** (Fig. 4a). At high Nd, LWP(on) − LWP(off) is +134 (40 km), +58 (5 km), +34 (2.5 km), +27 (1.25 km), +16 (625 m) and +6 $\mathrm{g\,m}^{-2}$ (312 m), all at Nd 1000. With suppression off, S is near zero or negative at every resolution.
3. **The falling branch comes from entrainment and is strengthened by sedimentation.** It survives with suppression off (at 312 m, $S_{100\to200}$ is -0.07 off and -0.05 on), while w_e keeps rising with Nd (Fig. 1). With suppression on it steepens with the sed factor (625 m: $S_{200\to1000} = -0.02/-0.04/-0.06$ for sed $\times$0.5/1/2; 312 m: $-0.02/-0.04/-0.07$ on and $-0.02/-0.05/-0.06$ off).
4. **Sedimentation–entrainment** (Fig. 5). At $\geq$10 km, w_e switches between two levels (about 4.0 and 3.0 $\mathrm{mm\,s}^{-1}$) at a cloud-top fall-speed proxy near 3.5. At $\leq$5 km, w_e falls smoothly with the proxy, and at a given proxy it is higher the finer the grid.
5. **The low-Nd branch is robust; LWP is not converged.** Below 5 km, $S_{20\to50}$ (suppression on) is 0.56–0.60, 0.65–0.76 and 0.88–1.00 for sed $\times$0.5/1/2, with little dependence on Δx. LWP, cloud cover and w_e still change strongly between 625 m and 312 m (Fig. 6): at Nd 100, LWP goes from 120 to 76 $\mathrm{g\,m}^{-2}$, cloud cover from 0.92 to 0.83 and w_e from 7.0 to 8.1 $\mathrm{mm\,s}^{-1}$, and the cells shrink (Fig. 7).

6. **3D turbulence makes almost no difference at 625 m or 312 m** (Fig. 9). At 625 m (36 pairs), $|\Delta\text{LWP}| < 1.5\,\text{g}\,\text{m}^{-2}$ and $|\Delta\text{cloud cover}| < 0.007$. At 312 m (16 pairs so far, Nd 20–200), $|\Delta\text{LWP}| \leq 1.0\,\text{g}\,\text{m}^{-2}$ and $|\Delta\text{cloud cover}| \leq 0.006$. The one consistent signal is w_e, which is 0.02–0.07 mm s^{-1} lower (<1 %).
7. **Steadiness** (Fig. 10). At ≤2.5 km, $|d\text{LWP}/dt| \leq 4.5\,\text{g}\,\text{m}^{-2}\,\text{h}^{-1}$ over hours 10–12. At 40–10 km, LWP is still rising at high Nd with suppression on, by up to $+18\,\text{g}\,\text{m}^{-2}\,\text{h}^{-1}$.
8. **Process-level diagnostics** (Section 3).
   - The LWP cell scale converges near 17–20 km (aspect ratio ≈17) by 625–312 m, even though LWP does not.
   - The resolved share of boundary-layer w-variance climbs from 0 at ≥10 km to 9 % (2.5 km), 37 % (625 m) and 66 % (312 m).
   - Drizzle decouples the boundary layer at low Nd (cumulus-like positive w skewness), and high Nd brings back Sc-like negative skewness, except at 312 m.
   - Precipitation suppression fades at fine resolution because thinner clouds make little rain and up to 90 % of it evaporates.
   - At fixed radiative driving, entrainment roughly doubles from 40 km to 625 m.
   - P3's `qc_sed`/`qr_sed`/`qi_sed` outputs are broken (a C++ port bug, diagnostics only).

Table 1: LWP (g m^{-2}, hours 10–12) with sed×1 and 3D turbulence off. Bold marks the maximum with suppression on.

Δx	suppression	Nd 20	50	100	200	500	1000
40 km	on	135	188	246	295	333	**342**
	off	234	221	216	211	209	208
10 km	on	134	186	240	281	308	**312**
	off	232	217	213	208	206	204
5 km	on	82	137	184	214	**222**	218
	off	135	149	153	158	159	160
2.5 km	on	67	119	163	**177**	175	170
	off	117	137	139	138	137	136
1.25 km	on	65	111	149	**157**	154	149
	off	116	127	127	124	122	122
625 m	on	52	98	120	**122**	119	115
	off	103	109	108	103	101	99
312 m	on	40	72	**76**	74	72	70
	off	79	79	72	69	66	64

1 Setup

Case and model. DYCOMS-II RF02 nocturnal drizzling stratocumulus (shortwave off; T, q nudged above 650 hPa; IOP surface fluxes and subsidence). The forcing is constant over 0–12 h. We run doubly periodic EAMxx (SHOC, P3, SPA, RRTMGP; 128 levels; stub sea ice). Domains are 600 km at 40 km, 300 km at 20 km, and 240 km from 10 km down. Time steps follow the earlier ACI study: the physics step is 100 s at ≥5 km and 75/60/30/15 s at 2.5/1.25/0.625/0.3125 km. The E3SM branch is `mahf708/eamxx/p3-cloud-sed-factor` (E3SM PR #9017 plus a P3 sedimentation knob).

Matrix. The full matrix is Δx (40, 20, 10, 5, 2.5, 1.25 km, 625, 312 m) × Nd target (20, 50, 100, 200, 500, 1000 cm^{-3}) × `cloud_sedimentation_factor` (0.5, 1, 2) × precipitation suppression (on, off), plus 3D turbulence (off, on) at 625 and 312 m.

- `p3::cloud_sedimentation_factor` is a new P3 option that scales the mass- and number-weighted cloud-droplet fall speeds, like MG2's `cld_sed`.
- Precipitation suppression off means KK2000 autoconversion with the Nd exponent set to 0 and the prefactor set to $1350 \cdot 63^{-1.79} = 0.812$. That matches the default rate at Nd 63.

- The SPA activation factor sets Nd. At ≥625 m the diagnosed in-cloud Nd is 10–25 % above target; at 312 m it is 15–20 % below.
- The 625 and 312 m runs are done in restart segments across interactive, debug and regular jobs. We checked that segmented runs are bit-for-bit identical to straight runs.

Diagnostics.

- Values are means over hours 10–12 unless stated otherwise. Cloud cover is the fraction of columns with LWP $> 10\,\mathrm{g\,m^{-2}}$.
- $w_e = dz_i/dt - w_{\mathrm{subs}}(z_i)$.
- $S = d\ln \mathrm{LWP}/d\ln N_d$, computed from the diagnosed Nd.
- The LWP spatial variance is computed online with the expression syntax, and cloud-top fields with the new `aerocom(..)` operator.
- The 6 h results (technical note v1) are consistent with the first 6 h of these runs, which reproduce them bit-for-bit.

2 Results

2.1 Nd response at every resolution

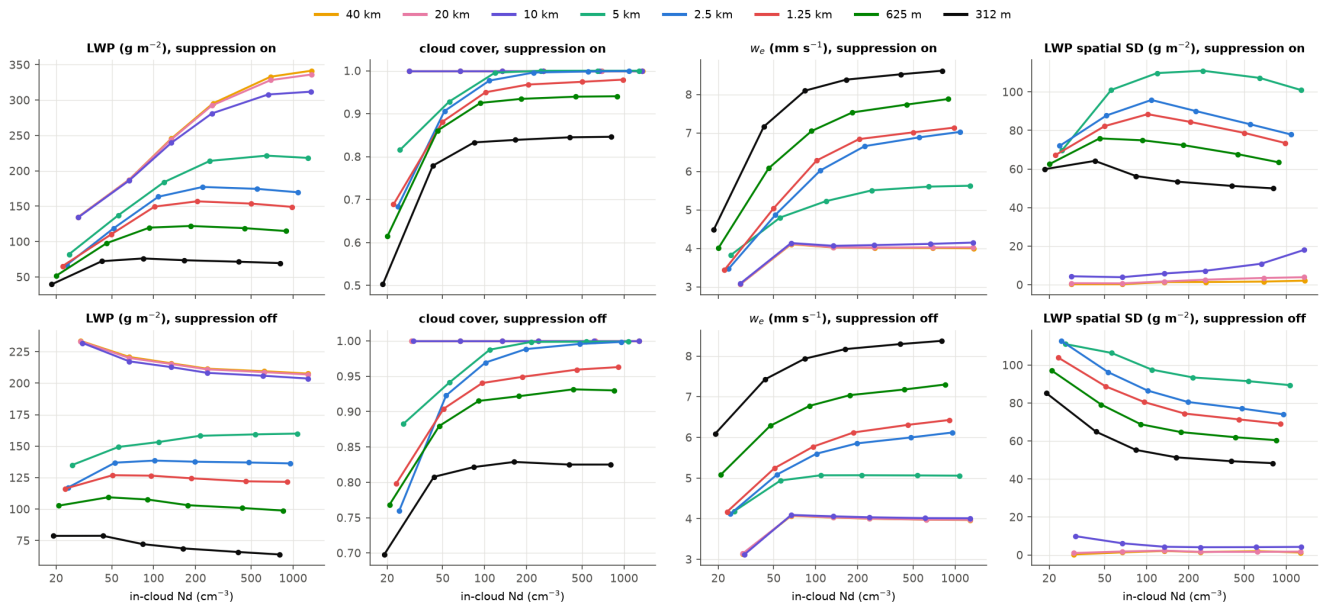


Figure 1: Nd response at sed×1, all resolutions on shared axes. Top row: suppression on. Bottom row: suppression off.

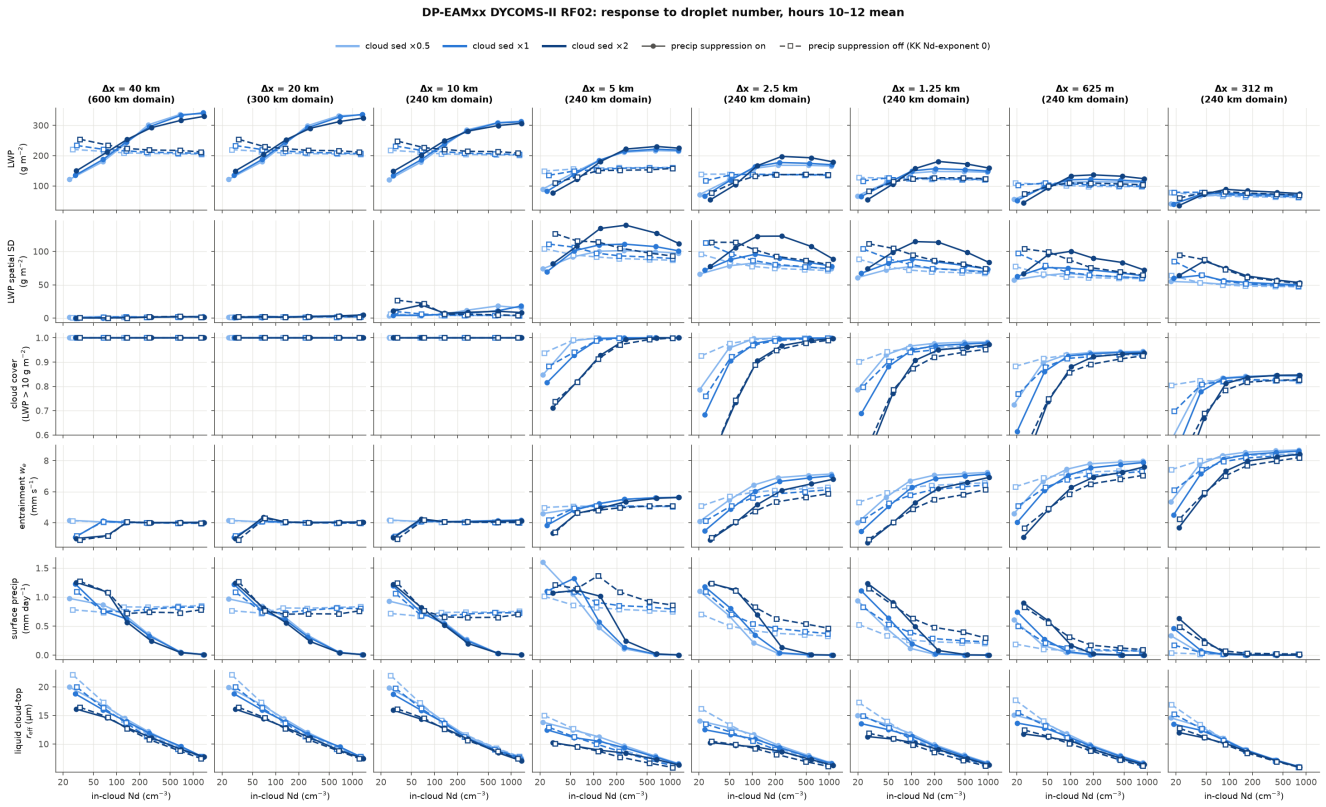


Figure 2: Every resolution, sed factor and suppression setting. Shade: sed factor. Solid lines: suppression on. Dashed lines: off.

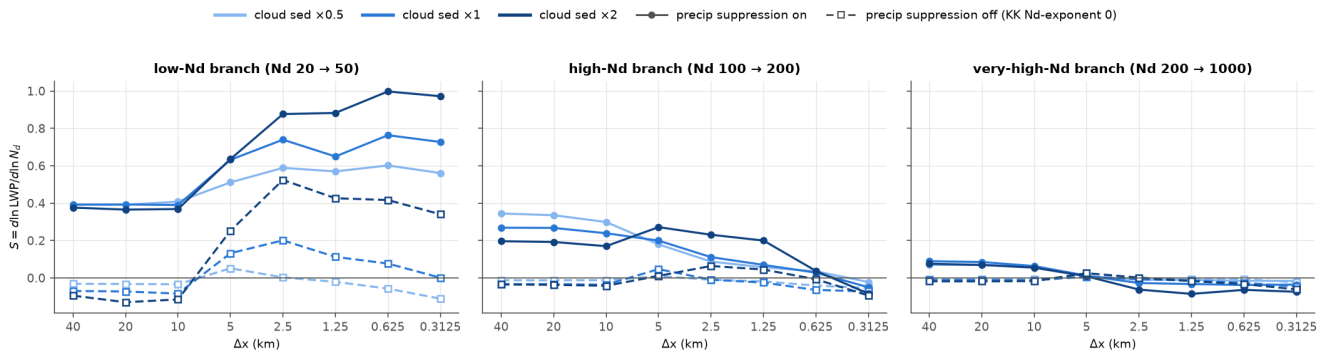


Figure 3: Susceptibility on three Nd branches against Δx.

2.2 Pathways

The pathway figure (Fig. 4) takes the difference between runs that differ in one switch only. The precipitation-suppression term crosses zero near Nd 63 by construction. Above Nd 100 it falls off steeply with Δx. The sedimentation term changes sign between the coarse and fine regimes.

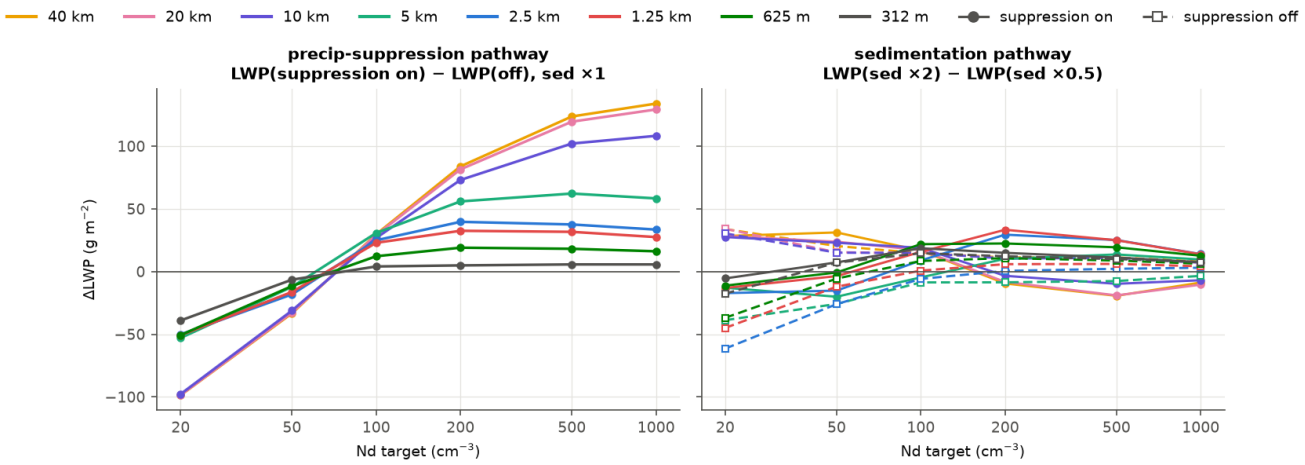


Figure 4: (a) Precipitation-suppression pathway (sed×1). (b) Sedimentation pathway, with suppression on (solid) and off (dashed). Colour: resolution.

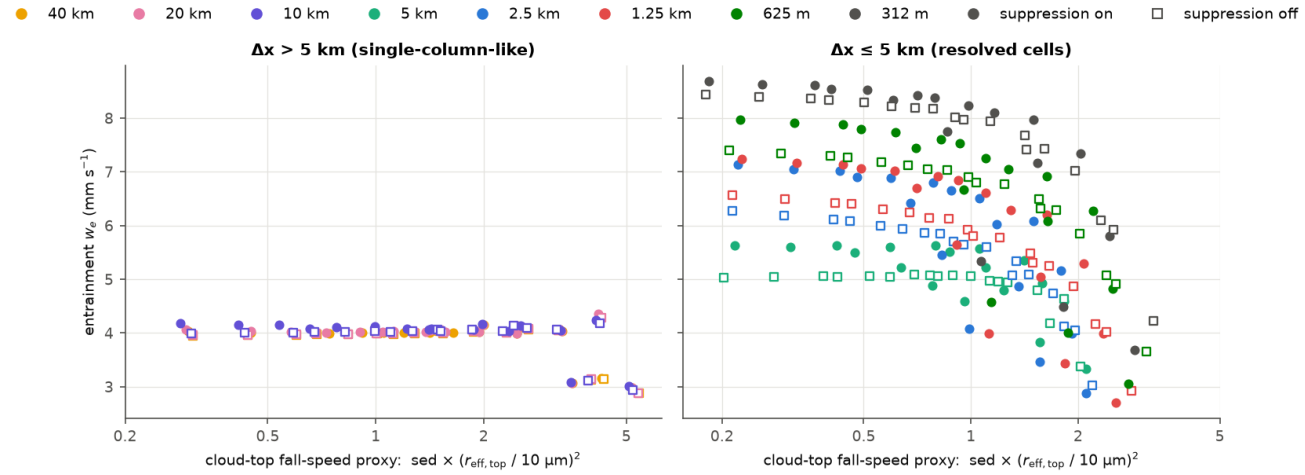


Figure 5: w_e against the cloud-top fall-speed proxy sed$\times(r_{\mathrm{eff,top}}/10\,\mu\mathrm{m})^2$, for every run with 3D turbulence off.

2.3 Resolution dependence and cloud morphology

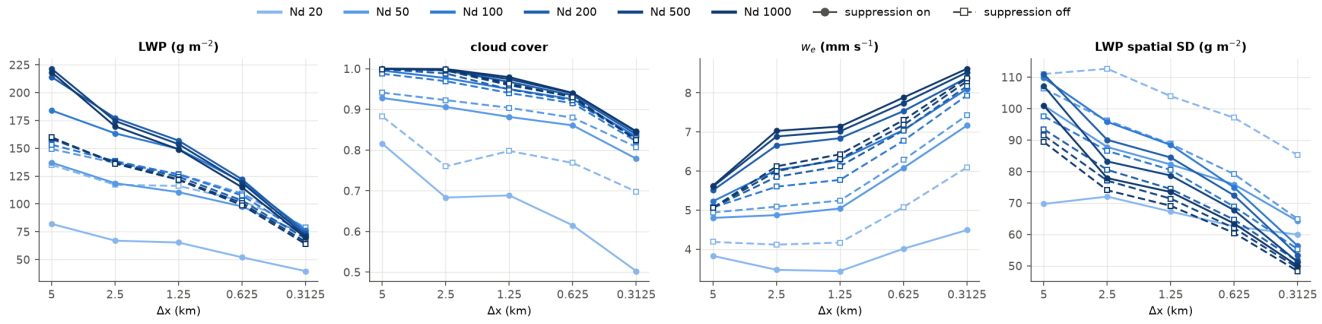


Figure 6: Hour 10–12 means against Δx ($\leq$5 km), sed×1, for every Nd.

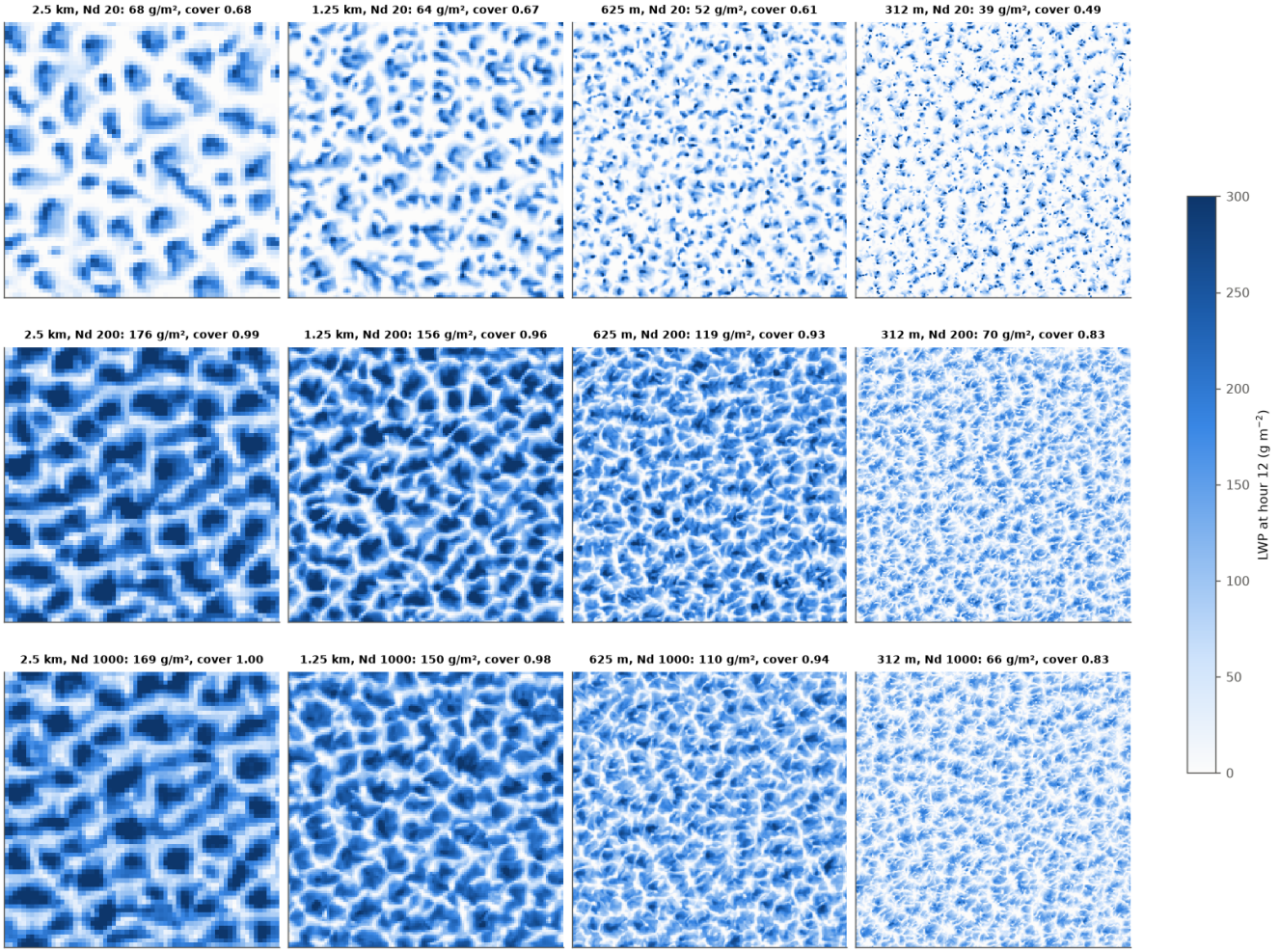


Figure 7: LWP at hour 12 over the full 240 km domain, sed×1, suppression on. Rows: Nd 20/200/1000. Columns: 2.5 km–312 m.

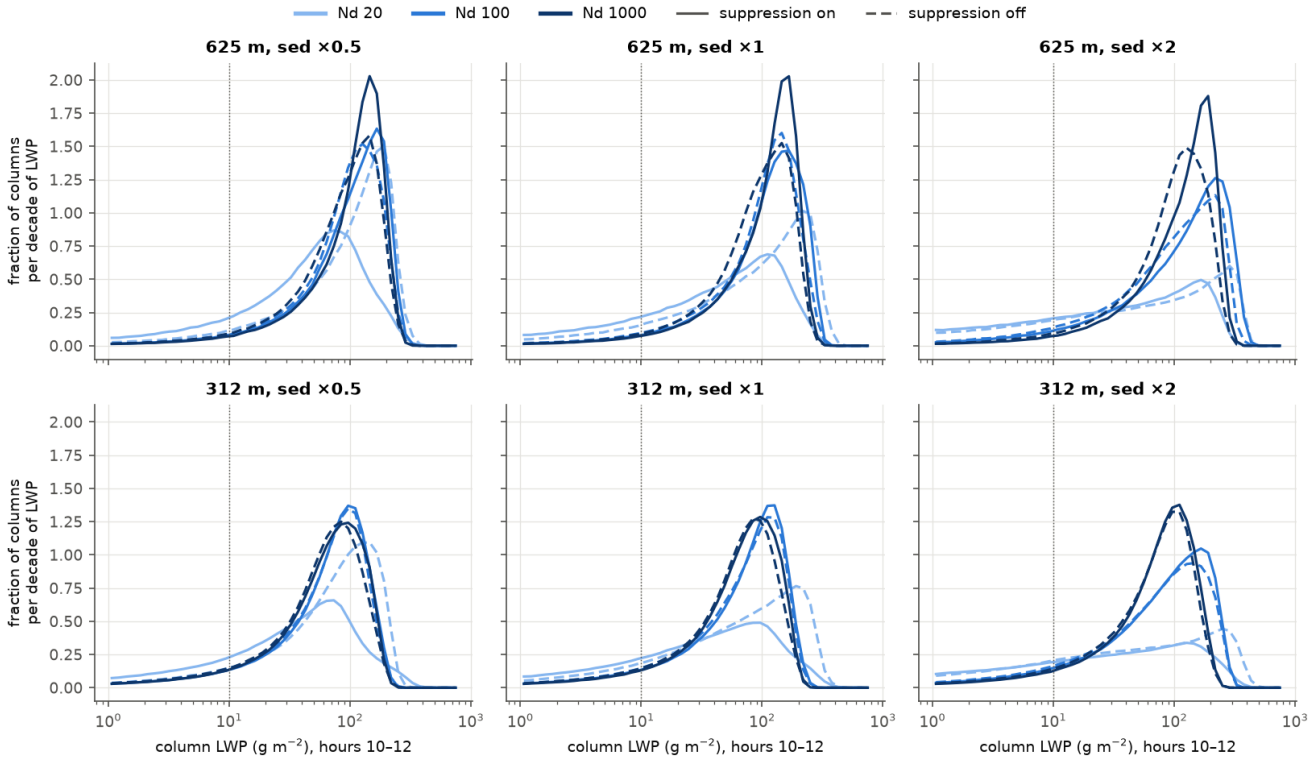


Figure 8: Column-LWP distributions over hours 10–12 at 625 m and 312 m.

2.4 3D turbulence

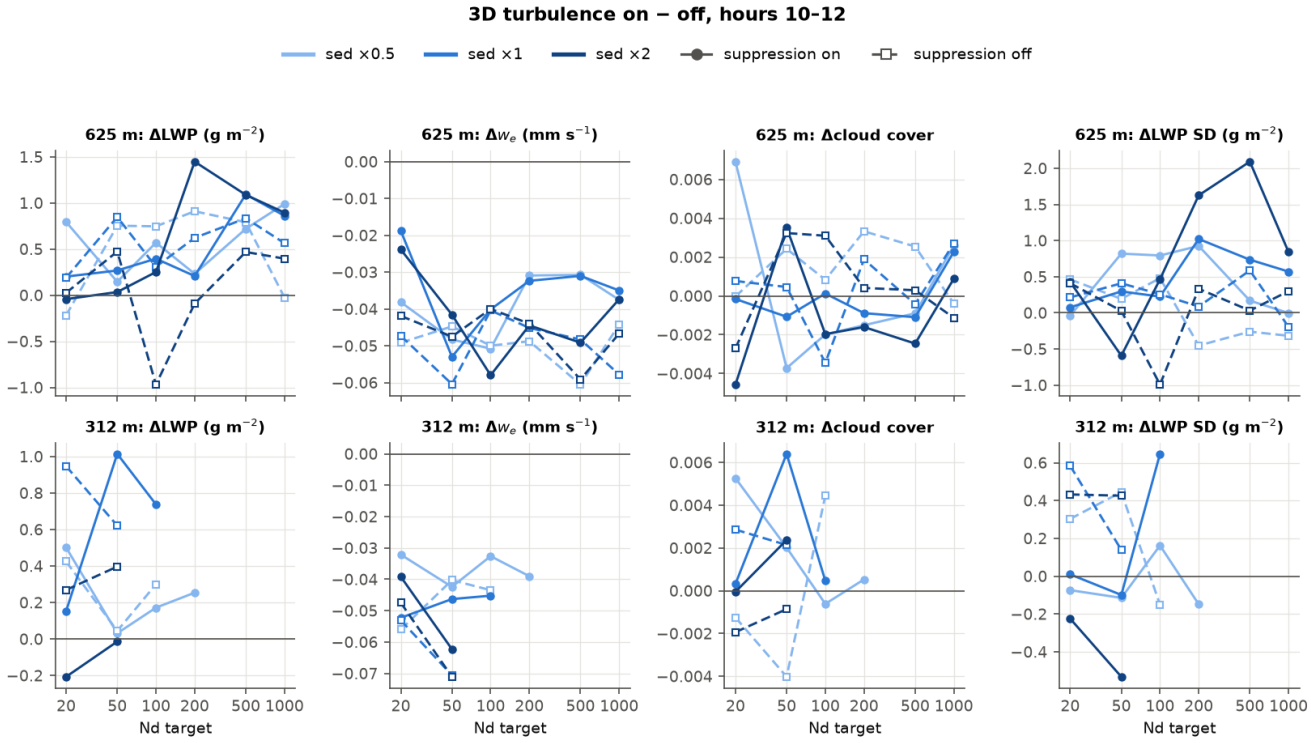


Figure 9: 3D turbulence on minus off at 625 m (all 36 pairs) and at 312 m (the 16 pairs complete so far).

2.5 Transience

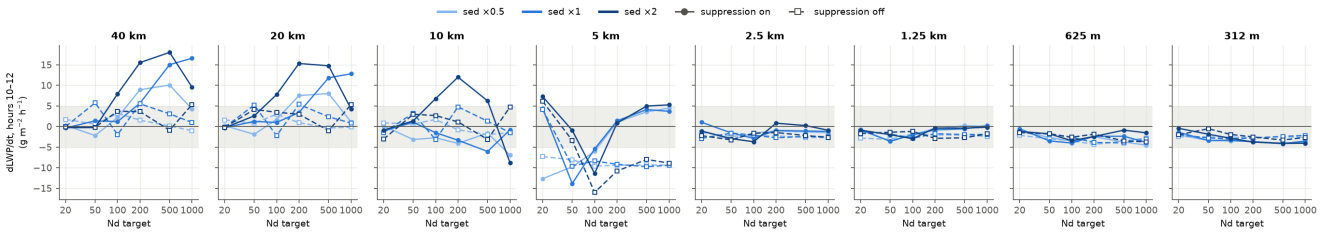


Figure 10: Linear LWP trend over hours 10–12. The grey band marks $\pm 5\,\mathrm{g\,m^{-2}\,h^{-1}}$.

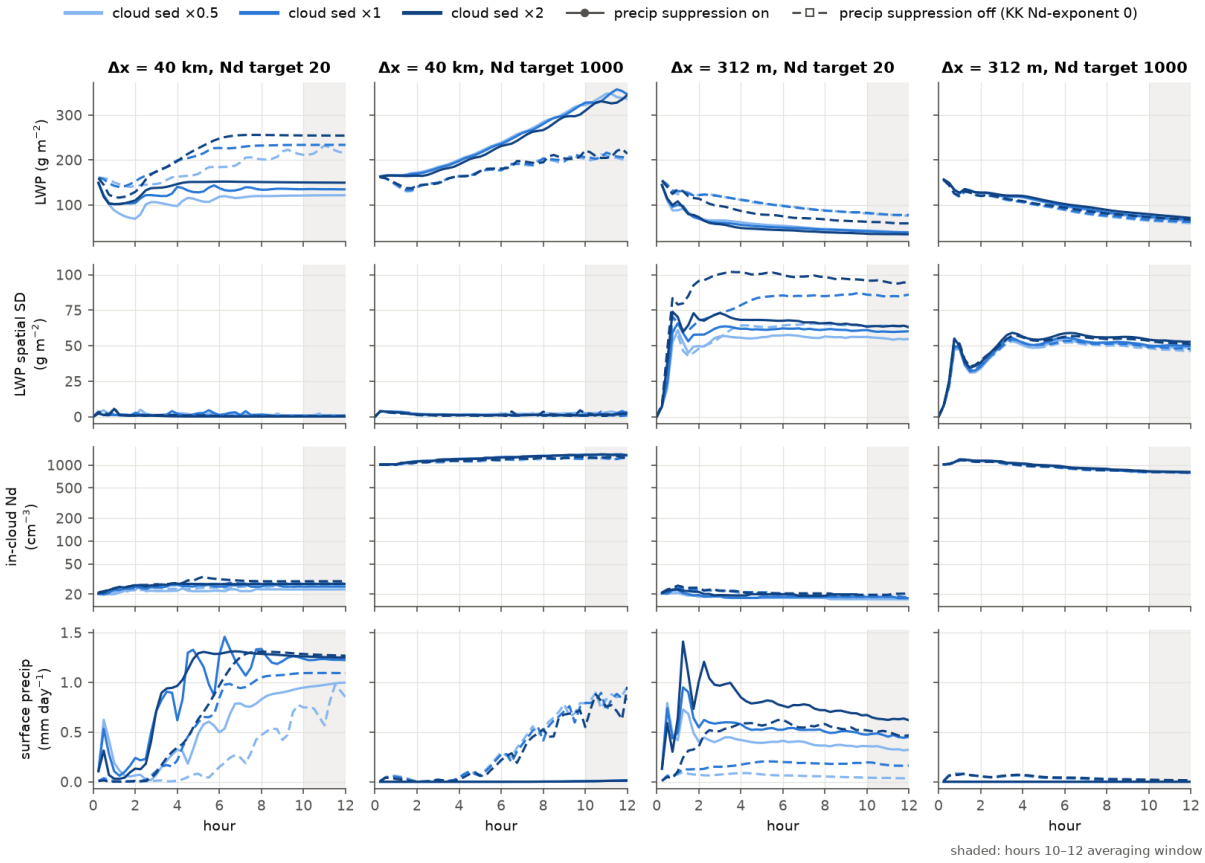


Figure 11: Time series at 40 km and 312 m for Nd 20 and 1000.

3 Process-level diagnostics

These come from the 15-min domain-mean profiles (hours 10–12), the 3D snapshots at hours 10 and 12 (for resolved w' and θ_v') and the 15-min column fields. All runs have 3D turbulence off. Resolved $w = -\omega/(\rho g)$. Decoupling follows Jones et al. (2011): Δq_t and $\Delta\theta_l$ between the lowest and highest quarter of the boundary layer. The cell scale λ is the peak of the radially averaged LWP power spectrum.

Cell scale converges; LWP does not (Figs. 12, 13). The dominant LWP wavelength falls from 30 km (2.5 km) to 24, 21 and 17 km (1.25 km, 625 m, 312 m). In grid units it rises from 8 to ≈37 physics cells, so below 1 km the cells are no longer grid-locked. The aspect ratio λ/z_i approaches ≈17–20, a typical value for observed mesoscale cells. At ≥5 km the domain sets the scale (48 km = 240 km/5). The spectra at 625 m and 312 m peak at nearly the same wavelength.

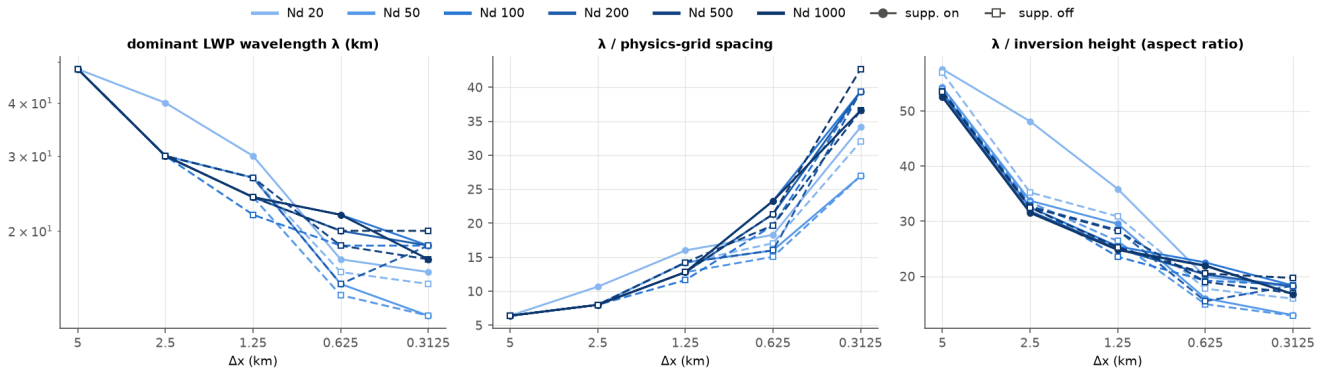


Figure 12: Dominant LWP wavelength, in km, in physics-grid spacings and relative to z_i, against Δx (sed×1).

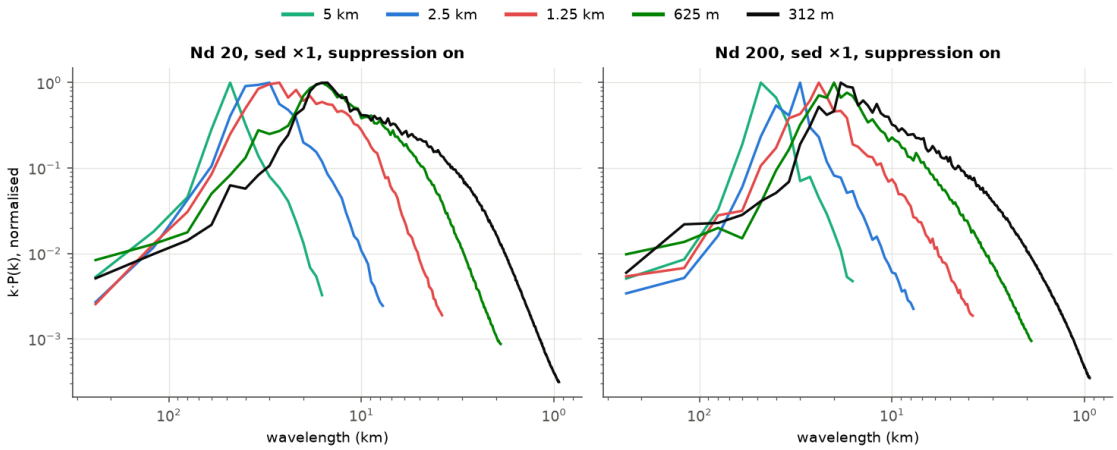


Figure 13: Radially averaged LWP spectra, $k\,P(k)$, normalised, hours 10–12.

The turbulence grey zone (Fig. 14). At ≥10 km all turbulence is SHOC's, and SHOC's boundary-layer w-variance is large (0.33–0.41 $\mathrm{m^2\,s^{-2}}$). Below 5 km SHOC's variance collapses to about 0.09 $\mathrm{m^2\,s^{-2}}$ and keeps falling with Δx, while resolved variance grows. Resolved motion makes up 6–9 % of the total at 5–2.5 km, 17 % at 1.25 km, 37 % at 625 m and 66 % at 312 m. Even at 312 m a third of the variance is still parameterised. Total variance is about 4× smaller at km scale than in the single-column runs, yet entrainment is higher. The resolved circulation entrains more efficiently than SHOC (see also Fig. 19).

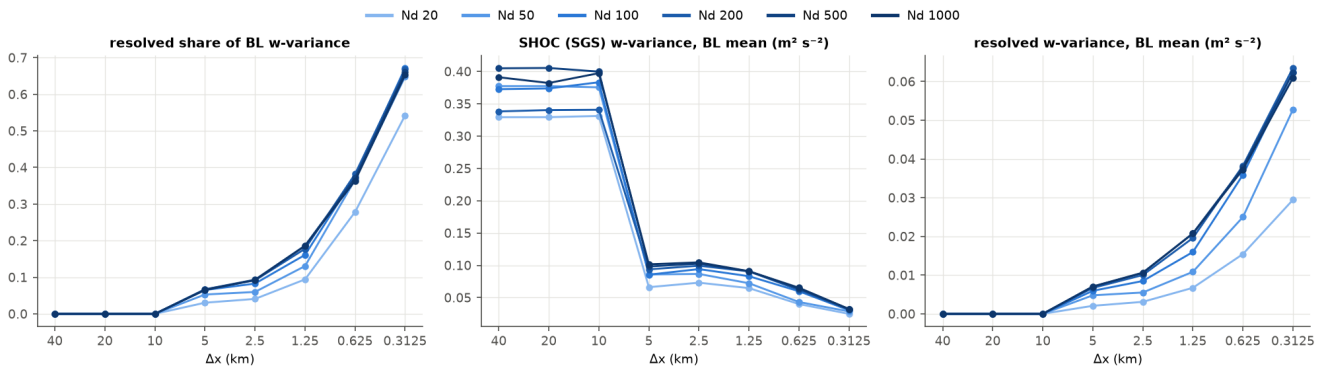


Figure 14: Resolved share of BL w-variance, SHOC w-variance and resolved w-variance against Δx (sed×1, suppression on).

Sc-like or cumulus-like turbulence (Fig. 15).

- At km scale, low Nd gives positive resolved w skewness (up to 1–2): narrow, strong updrafts, a cumulus-like, drizzle-decoupled boundary layer.
- High Nd gives negative skewness (-0.6 to -0.95 at 2.5 km–625 m), the radiatively driven downdraft signature of a well-mixed Sc layer.
- The switch happens near Nd 50–100 and moves to higher Nd with more sedimentation.
- At 312 m the skewness stays positive at every Nd (0.1–0.3 for Nd ≥ 100), so its cloud-topped boundary layer looks structurally different. That is a convergence question for 156 m.
- With precipitation suppression off the skewness depends less on Nd.

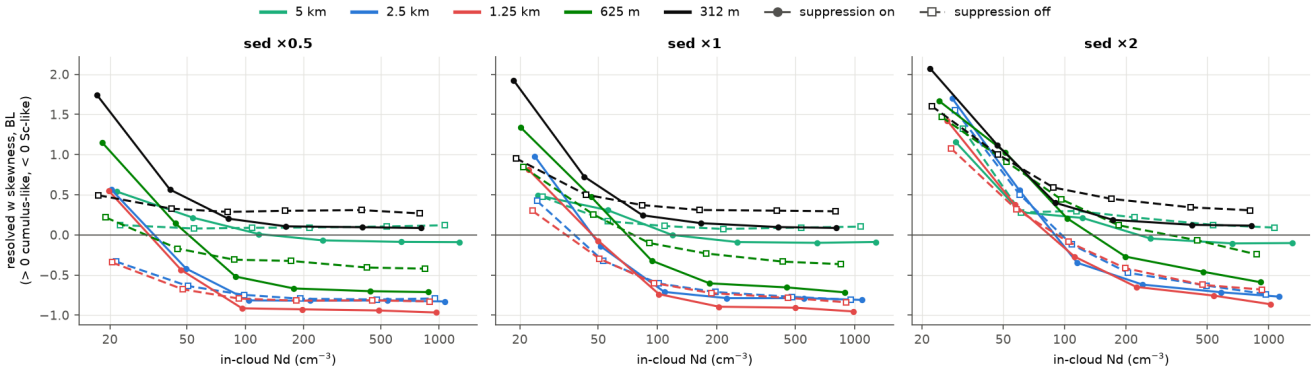


Figure 15: Resolved w skewness over the boundary layer against Nd, for each sed factor ($\Delta x \leq 5$ km).

Drizzle-induced decoupling (Fig. 16). At ≤ 5 km, Nd 20 has $\Delta q_t \approx 1.1$–$1.6\,\mathrm{g\,kg^{-1}}$ and $\Delta\theta_l \approx 1.4$–$2.2$ K, which is strongly decoupled. Both fall steeply with Nd, to 0.2–$0.6\,\mathrm{g\,kg^{-1}}$ and 0–0.7 K by Nd 1000. This is the classic picture: below-cloud evaporation of drizzle stabilises the subcloud layer. In the single-column runs, q_t decreases upward while θ_l is unstable for Nd ≥ 50 ($\Delta\theta_l < 0$) across a deep SHOC-mixed cloud layer that starts at about 240 m (Fig. 18). That is a structurally different boundary layer, not a coarse version of the km-scale one.

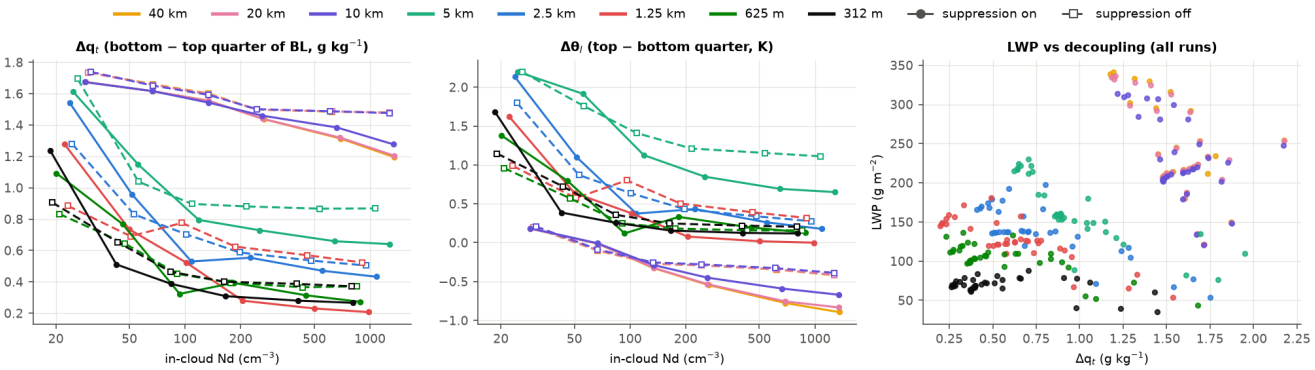


Figure 16: Decoupling: Δq_t and $\Delta\theta_l$ against Nd (sed×1), and LWP against Δq_t for all runs.

Why precipitation suppression fades with resolution (Fig. 17). At fixed Nd, column rain production (autoconversion plus accretion) falls steeply with Δx. At Nd 200 it goes from 0.5 (40 km) to $0.03\,\mathrm{mm\,d^{-1}}$ (312 m), because the clouds get thinner: cloud thickness falls from 600 m to 330 m and cloud base rises from 240 to 680 m. The share of that production that evaporates before reaching the surface rises from about 0.4 to 0.93. The raining fraction of the domain drops to zero above Nd 100 at ≤ 1.25 km. With almost no rain to suppress, the suppression pathway has nothing to act on.

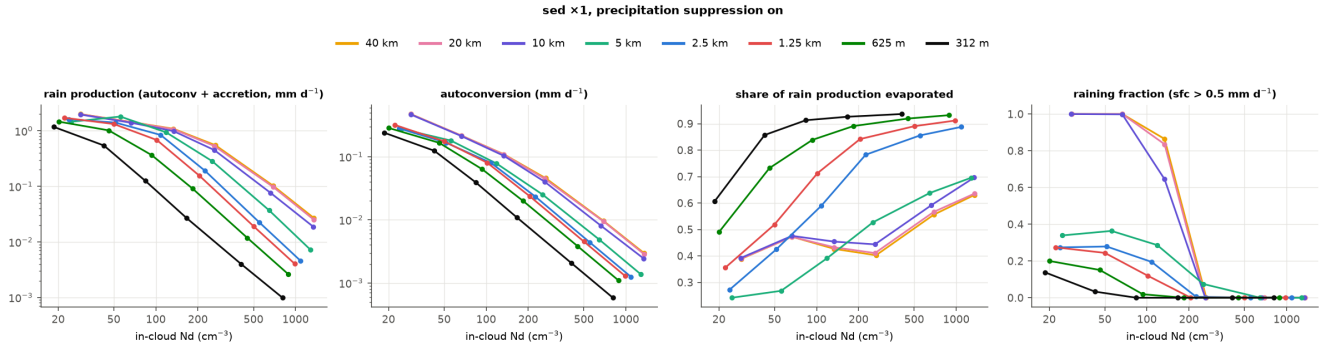


Figure 17: Column rain production, autoconversion, evaporated share and raining fraction against Nd (sed×1, suppression on).

Mean structure (Fig. 18). As Δx shrinks, the boundary layer deepens (z_i: 860 m at 40 km to 1010 m at 312 m), warms and dries in the mixed layer, and the cloud layer thins and lifts. LW cooling at cloud top is sharper and stronger in the single-column runs.

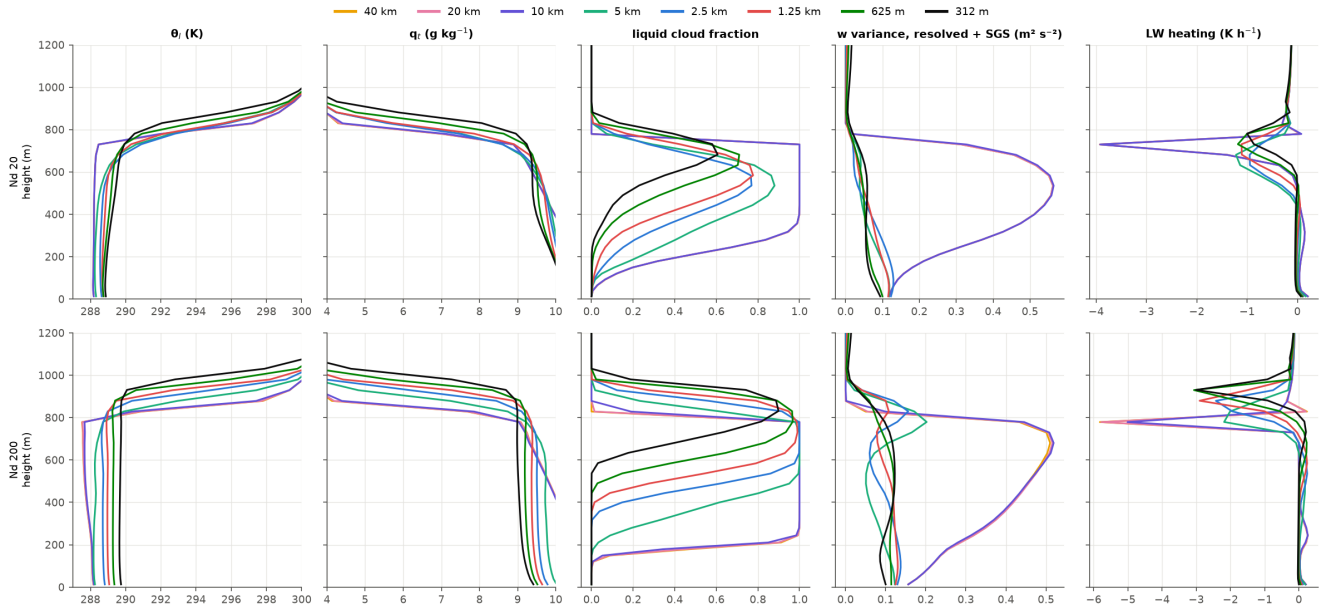


Figure 18: Hour 10–12 mean profiles at Nd 20 (top) and 200 (bottom), sed×1, suppression on. The 40, 20 and 10 km lines overlap.

Radiative driving and entrainment efficiency (Fig. 19). BL LW cooling saturates at $\approx$86 W m^{-2} for LWP $>$ 120 g m^{-2}, so most runs share almost the same radiative forcing. At that forcing, the median w_e is 4.0 (40 km), 5.1 (5 km), 6.1 (2.5 km), 6.6 (1.25 km) and 7.8 mm s^{-1} (625 m). Entrainment per unit radiative driving therefore roughly doubles from the single-column limit to 625 m. This is a resolution effect on the entrainment closure, not on the forcing.

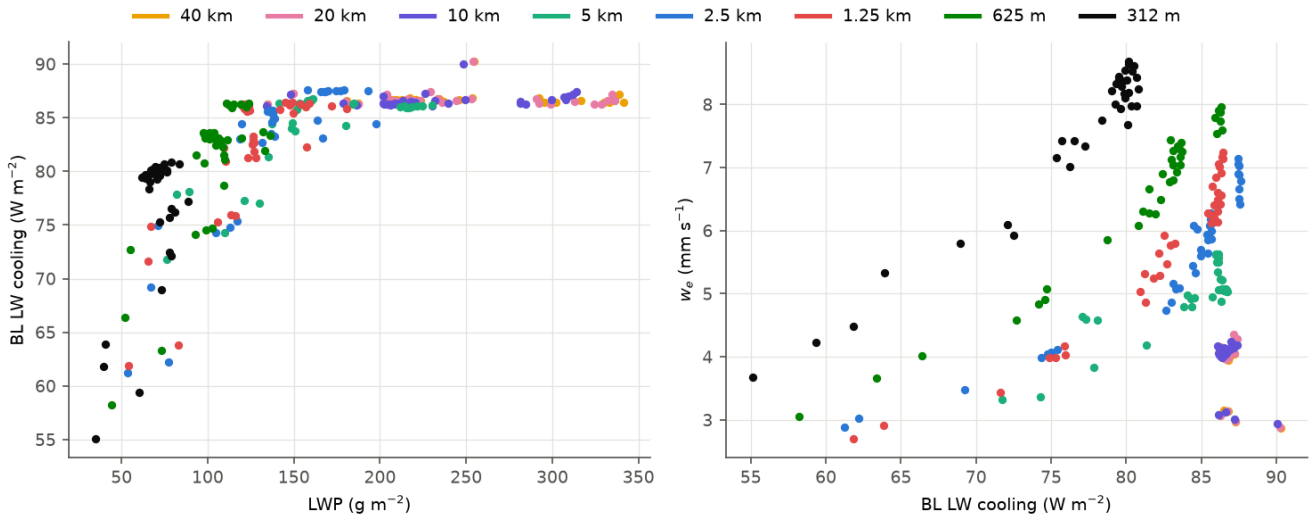


Figure 19: BL LW cooling against LWP, and w_e against BL LW cooling, for all runs.

A diagnostic bug in P3. P3's sedimentation tendency outputs `qc_sed`, `qr_sed` and `qi_sed` cannot be used. In the C++ port, `cloud_/rain_/ice_sedimentation` compute `q_tend = (q - q_tend)/dt` and expect `q_tend` to hold the pre-sedimentation q. `p3_main` never loads it (the Fortran sets `qc_sed = qc` first), so the outputs come out as $\approx q/\Delta t$. For example, `qc_sed/qc` $= 1/(60\,\mathrm{s})$ at 1.25 km. Model state is not affected. The fix is to copy `qc`, `qr`, `qi` into the three buffers before the sedimentation calls. This also explains the "nonsense" `qc_sed` noted in the earlier ACI study.

4 Caveats, status and next steps

- **312 m is 76 % done** (658 of 864 model-hours).
  - Every run with 3D turbulence off is complete. Of the 36 runs with it on, 20 are still to come.
  - Perlmutter maintenance runs from 11 to 18 November. Regular jobs sized to finish before it are queued; anything unfinished resumes afterwards from its last restart.
  - A final version of this report will follow.
- **No convergence.** The step from 625 m to 312 m is as large as any coarser step. 156 m (or a smaller domain at 156 m) is the open question for discussion.
- **Single realization per run.** Seed ensembles at the km-scale and sub-km high-Nd runs would put error bars on the small negative S values.
- **Diagnostics.**
  - Surface precipitation includes cloud-droplet sedimentation.
  - w_e comes from a linear fit of z_i.
  - The fall-speed proxy stands in for the actual cloud-top sedimentation flux.

5 Data and code (Perlmutter, world-readable)

- 12 h runs: `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/sedps_<res>_nd<Nd>_sed<0p5|1|2>_ps<on|off>[_t3don]/run/`
- This report's analysis: `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/analysis_report4/`
  - `summary_h10-12.csv` and `timeseries_15min.csv`: one row per run (and per 15-min step), with columns for the run settings including `turb3d`.
  - `susceptibility.csv`, `lwp_trend_h10-12.csv`, `turb3d_on_minus_off.csv` and all figures.

- 6 h runs and technical note v1: `/pscratch/sd/m/mahf708/dp_screamxx_sedps/`
- Scripts, R tools and LaTeX source: `/pscratch/sd/m/mahf708/scmlib/sed-precsupp-study/`
  - In R: `source("../R/sedps.R"); sedps_runs("/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h", hours = 12)`.
- Code: E3SM `mahf708/eamxx/p3-cloud-sed-factor` and scmlib `sed-precsupp-study`.